

A Packing of 60 Lines in $\mathbb{R}^{14}$

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Abstract

We give 60 1-dimensional subspaces of $\mathbb{R}^{14}$ whose smallest squared chordal distance is at least 0.920334. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.919922, obtained by deleting 5 of the 65 lines of the packing listed in Sloane's 1997 table. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 60 subspaces of dimension 1 in $\mathbb{R}^{14}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.920334 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 14$, $n = 1$. The cell $N = 60$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.919922, obtained by deleting 5 of the 65 lines of the packing listed in Sloane's 1997 table. The Rankin simplex bound for this cell is $1(14 - 1)/14 \cdot 60/59 = 0.944310$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m14n1N60.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 70 seconds on one laptop, with about ten such searches running at once.

A Exact value

```
numerator:
230083575798324906224049775244338007235447277811
denominator:
249999999999840226399372122655299971323174914180
```

B Coordinates

60 blocks of 14 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m14n1N60.txt`.

-141469518331	175155987316	55576745842	-117581079712
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-129762148115	3948168982	-351911911200	277683684394
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